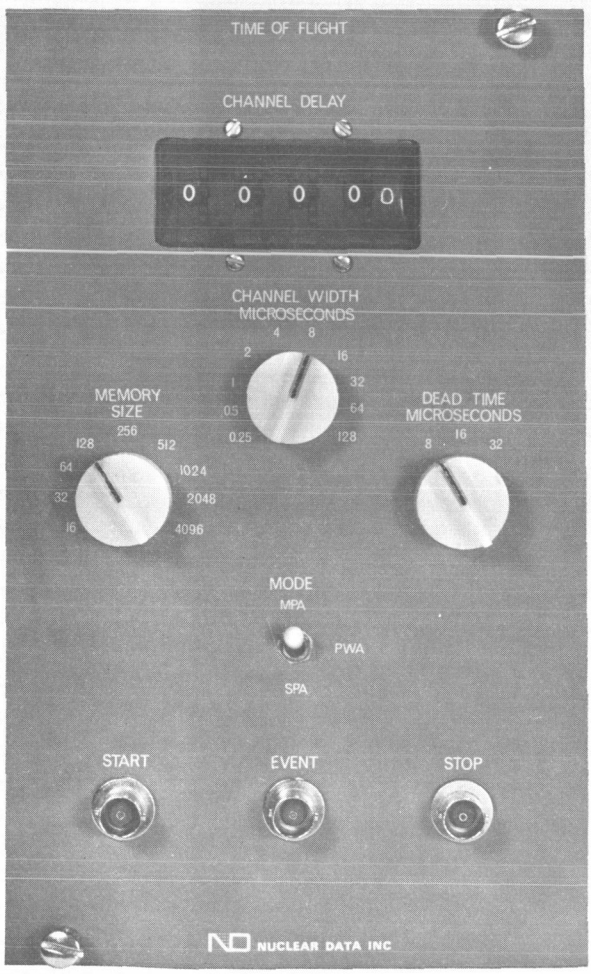
TECHNICAL SPECIFICATIONS

NUCLEAR DATA INC
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SERIES 2200 OR 3300 TIME OF FLIGHT

This NIM compatible Digital Time of Flight Module is a highly accurate and versatile unit that can be used with the Series 3300 or 2200 System Analyzer (as specified when ordered) to measure one-quarter microsecond minimum channel widths when performing operations such as encountered in neutron time of flight measurements. Pulse width analysis and single pulse analysis experiments may also be performed as selected by a front panel switch. A 32 MHz, crystal controlled oscillator minimizes channel time jitter and insures long term stability. A switch selectable, delay time control is included that incorporates no fixed internal delay and can be used to delay the start of analysis for up to 99,990 channels. A variable dead time, and memory capacity switch completes the complement of controls.

Start Input

Pulse Requirements
- Amplitude: 2 to 20 volts.
- Polarity: Positive.
- Rise Time: Noncritical; should be uniform.

Input Impedance — Approximately 2000 ohms.

Event Input

Pulse Requirements
- Amplitude: 2 to 20 volts.
- Polarity: Positive.
- Rise Time: Noncritical; should be uniform.

Input Impedance — Approximately 2000 ohms.

Stop Input

Pulse Requirements
- Amplitude: 2 to 20 volts.
- Polarity: Positive.
- Rise Time: Noncritical; should be uniform.

Input Impedance — Approximately 2000 ohms.

Input Signal Attenuation

None required as long as signal returns to less than one volt prior to completion of operation.

Clock Oscillator

32 MHz, crystal-controlled.

Channel Width (Time Duration)

0.25 to 128 microseconds switch selectable in binary increments.

Channel Time Jitter

$\pm 1/64$ microsecond.

Delay Time

0 to 99,990 channels thumbwheel switch selectable in 10 channel increments.

Fixed Delay

No fixed delay is incorporated in this unit. Delay times are as selected.

Dead Time

8, 16 or 32 microseconds, switch selectable.

Memory Size

16 to 4,096. Channels switch selectable in binary increments. 8192 or 16,384 sizes optional.

Power Requirements

+12 volts D.C. at 1 ampere.
−12 volts D.C. at 250 milliamperes.

Size

8.71" high x 5.36" wide x 9.7" deep (equivalent to four standard AEC single width modules).

Part Number

Series 2200; 74-0145
Series 3300; 74-0146

SPECIFICATIONS SUBJECT TO CHANGE WITHOUT NOTICE
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